AIRSTREET · PRODUCT REQUIREMENTS AND SPECIFICATIONS

AirStreet product requirements and specifications

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DOCUMENT CONTROL

Document ID	AST-PRS-001	Version	0.9
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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/airstreet		
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REVISION HISTORY

Version	Date	Author	Change
0.9	2026-10-24	Amish Chadha	Kit 1.9.1 upgrade: AST-REQ-001 becomes AST-PRS-001; R1 to R16 become PR-001 to PR-016, each traced to a user requirement (AST-URS-001) with target, tolerance and units, the old number under Was; STEM check fixes of AST-DDR-005 (rated-range flags in PR-007, NO2 noise and ozone filter in AST-CAL-001 v0.7); PR-013 3.77 kg with the hollow shield spacers; PR-017 and PR-018 added for UR-014 and UR-015, proposed, awaiting Amish
0.8	2026-10-02	Amish Chadha	Status from AST-CAL-001 v0.6 with the four lightening steps carried into the design: R13 at risk to met on paper (3.78 kg with the sun shield); R15 over the target by $17.50; R5 figures updated
0.7	2026-10-02	Amish Chadha	R13 mass limit set at 4.0 kg including FieldNode's sun shield (AST-DDR-003

			A1, accepted 2026-10-02); R13 now at risk
0.6	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.5	2026-09-30	Amish Chadha	Status from AST-CAL-001 v0.3 after the design was made constructable (AST-DDR-003); R13 and R15 not met, proposed to Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 status from AST-CAL-001; R15 redefined as the sensor-head cost (AST-DDR-001 D1); R6, R8 and R11 rest on adopted choices
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

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What the node must do, each requirement traced to a user need in AST-URS-001 (links in AST-RTM-001, `03b-traceability.md`). Until kit 1.8 this document was AST-REQ-001; R1 to R16 are now PR-001 to PR-016 with the same text and targets, and the old number is in the "Was" column, which the calculation note AST-CAL-001 and earlier records still cite. The requirements are checked by calculation in AST-CAL-001 v0.7 and by the STEM check AST-STM-001; targets remain proposals for review with the first co-design partner.

On 2026-09-25 Amish accepted the recommendations in AST-DDR-001 and AST-DDR-002: PR-003 (was R3) is withdrawn and kept as a research question, PR-008, PR-012, PR-013, PR-015 and PR-016 are restated, and PR-015 uses the $285 `budget_usd`. On 2026-09-30 the design was made constructable (AST-DDR-003); on 2026-10-02 Amish set PR-013's mass limit at 4.0 kg including FieldNode's sun shield, with the four lightening steps carried into the design. On 2026-10-24 the STEM check found four details to fix (AST-DDR-005, made under Amish's standing instruction of 2026-10-23): the NO2 noise term now follows the sensor datasheet, the NO2 cell is replaced before its ozone filter is used up, every record carries rated-range flags (PR-007), and the band thickness in the model matches the calculation; the hollow shield spacers of the D3 model take 0.01 kg off (PR-013 now 3.77 kg). The independent review of the same day corrected the NO2 budget (hourly MAE about 4.3 ppb, from 3.9) and the EU limit band; no status changed. Status: five at risk, five met on paper, six met by design (PR-017 and PR-018 among them, proposed), one over the value-engineering target and one withdrawn; none is not met.

Table 1. Product requirements and specifications. Status at TRL 3 from AST-CAL-001 v0.7 and AST-STM-001 v0.1.

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
PR-001	UR-001	Performance	PM2.5 at street level	The node shall measure PM2.5 over 0 to 500 ug/m3 with an hourly mean, after humidity correction, within the tolerance of a collocated reference	0 to 500	±5 ug/m3 or ±30 %, whichever is larger	ug/m3	Analysis (AST-CAL-001 E), TRL 3; CalRig chamber check and field collocation, TRL 4	At risk: wi the bound to 80 % R outside it 85 % RH a above, wh water upta makes the correction uncertain
PR-002	UR-002	Performance	NO2 at street level	The node shall measure NO2 over 0 to 200 ppb (about 0 to 380 ug/m3) with an hourly mean absolute error after field calibration against a reference station	5	max	ppb (hourly MAE)	Analysis (AST-CAL-001 C), TRL 3; collocation with a chemiluminescence reference, TRL 4	At risk: ab 4.3 ppb fr a budget c literature assumed terms, wit the datash noise over independe records ar hour and a % slope te (AST-DDR-005 and the independe review); h outside th sensor's r range are flagged (PR-007)
PR-003	UR-003	Performance	NO2 near the WHO annual guideline	Withdrawn as a requirement (AST-DDR-002). Research question: can the annual mean be resolved to 2 ug/m3 at 10 ug/m3?	2	max	ug/m3 (annual, 1 sigma)	Collocation statistics (research)	Withdrawn annual uncertaint about 4.6 ug/m3 (1 sigma); no WHO guideline claim
PR-004	UR-003	Performance	Streets above the EU NO2 limit value	The data shall classify a street's annual NO2 mean above or below	40	90 % one-sided confidence	ug/m3	Analysis (AST-CAL-001 C), TRL 3; collocation statistics, TRL 4	At risk: decisive o below 33. above 46. ug/m3 (at

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
				40 ug/m3 with 90 % confidence					directive's degC). The recast directive (2024/288 sets 20 ug from 2030 (AST-REG-001 with a ban 13.9 to 26 ug/m3; a target is proposed, awaiting Amish (AS DEC-001)
PR-005	UR-001, UR-002	Performance	Temperature and humidity for corrections	The node shall measure air temperature and RH inside a radiation shield for the PM and NO2 corrections	±0.5 degC, ±3 % RH; radiation error 1 degC or less in full sun at 1 m/s	max	degC; % RH	Analysis (AST-CAL-001 D), TRL 3; CalRig and side-by-side with an aspirated reference, TRL 4	At risk: SH meets the sensor figures; sh error 0.42 0.87 K at s, which re 4.1 % RH at 85 % R
PR-006	UR-005	Performance	Stated calibration method	T, RH and PM shall be checked on CalRig before deployment; NO2 shall be collocated with a reference station for at least 14 days before deployment and at least every 6 months	14 days; 6 months	min; max	days; months	Inspection of the calibration record per node, TRL 4	Met by de (AST-DDR-001
PR-007	UR-006	Software	Raw data for recalibration	The node shall send raw NO2 working and auxiliary electrode signals, PM, T and RH with a calibration	All raw signals, version tag, range flags	Pass or fail	n/a	Inspection of the firmware and data schema, TRL 4	Met by de flags set w the shield is above 3 degC, RH above 80 (PM) or 85 (NO2) or

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
				version tag, and flags for readings outside the sensors' rated temperature and humidity ranges (AST-DDR-005 F2)					below 20 (PM) or 15 (NO2), or below 10 degC (PM)
PR-008	UR-004	Software	Reporting for street maps	One record every 5 min through a private gateway (TwinKit by default), one every 15 min on The Things Network; hourly means published within 15 min; 7 days of store and forward	5 (private), 15 (TTN); 15; 7	max; max; min	min; min; days	Analysis of timing, airtime and storage (AST-CAL-001 B), TRL 3	Met on pa at 15 min fair use is up to SF9; days in 40 kB
PR-009	UR-013	Electrical	FieldNode power budget	The sensor load shall fit FieldNode's allowance and be energy neutral at FieldNode's design sun hours	115 (ceiling); 100 (design)	max	mW	Power budget (AST-CAL-001 A), TRL 3; energy log, TRL 4	Met on pa 87.0 mW design loa 2.33 Wh/d against 7. Wh stored the worst month; 6. days with sun
PR-010	UR-016	Electrical	Radio duty cycle	LoRaWAN airtime shall stay below 1 % in each EU868 sub-band, and within the 400 ms dwell time in US915	1; 400	max	%; ms	Airtime calculation (AST-CAL-001 B), TRL 3	Met on pa 0.082 % a SF9, 0.60 at SF12; 2 ms at SF9 US915
PR-011	UR-007	Physical architecture	Pole mounting without drilling	The node shall fit poles 80 to 200 mm in diameter with band clamps only, inlets	80 to 200; 1.5 to 4	range	mm; m	Design review and model (AST-CAL-001 F), TRL 3	Met by de inlet 3.0 m (AST-DDR-001 140 degre saddles ar

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
				between 1.5 and 4 m above ground					bands cut length
PR-012	UR-009	Physical architecture	Survive outdoors	Pod inlets face down behind insect mesh under a drip lid; electronics in FieldNode's IP65 enclosure; survive -10 to 50 degC, with FieldNode's sun shield at sites above 45 degC	-10 to 50	range	degC	Design review (AST-CAL-001 H), TRL 3; field trial, TRL 4	At risk: th NO2 cell i rated to 4 degC and % RH and SPS30 is recomme to 40 deg and 80 % so hours beyond ar flagged (PR-007); condensa and insect unverified
PR-013	UR-008	Physical architecture	Light and compact	Mass on the pole including FieldNode's sun shield, and frontal area seen from the street	4.0; 0.15	max	kg; m2	Constructable model (AST-CAL-001 F), TRL 3; weighing, TRL 4	Met on pa 3.77 kg w the sun sh (3.61 kg without), kg margin 0.139 m2 pole owne confirmat of the loa still to co
PR-014	UR-011	Software	Levels only	No images, audio or personal identifiers shall be collected or sent	None	Pass or fail	n/a	Design review, TRL 3	Met by de
PR-015	UR-012	Cost	Low cost	AirStreet sensor-head parts at or under the $285 value-engineering target per node (`budget_usd` , a hypothetical control target); FieldNode core	285 (value-engineering target)	max	USD	Priced BOM (AST-CAL-001 I), TRL 3	**Over the value-engineeri target by $17.50**: sensor he $302.50; $433.50 the FieldN core

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
				excluded and costed in FieldNode					
PR-016	UR-010	Maintenance	Serviceable in the field	A pre-collocated sensor pod shall be exchanged at the pole with hand tools; no sensor-level swaps at the pole (AST-DDR-002); the NO2 cell is replaced on the ground at the pod exchange every 12 months, or 18 or 24 where the site's ozone record keeps its filter under 500 ppm h (AST-DDR-005 F3)	15	max	min	Service time estimate (AST-CAL-001 H), TRL 3; timed exchange, TRL 4	Met on pa about 14 for a pod exchange
PR-017	UR-014	Materials	Open and garage buildable	Every made part shall be printable in ASA on a 250 mm class printer, or cut and drilled from standard aluminium bar, angle and sheet with hand tools and a drill; bought parts are specified by class, not brand	All made parts by printing, sawing and drilling	Pass or fail	n/a	Inspection of the build plan AST-BLD-001 and the drawings, TRL 3	Met by de (text proposed, awaiting Amish)
PR-018	UR-015	Safety	Installation and service safety	The node shall be installed and serviced only with the pole owner's written permission, from a work platform or	Owner's permission; no pole access; load stated	Pass or fail	n/a	Inspection of the build plan safety stops and AST-RSK-001, TRL 3	Met by de (text proposed, awaiting Amish)

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
				secured ladder with traffic management, without opening the pole; mass and wind load are stated for the owner's check; the FieldNode cell is fused and protected					

Categories: physical architecture, performance, materials, usability, packaging and transport, safety, electrical, software. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Assumptions

- NO2 conversions use 1 ppb = 1.88 ug/m3 at 25 degC and 101.325 kPa.
- The WHO guideline level for annual NO2 is 10 ug/m3 and for PM2.5 is 5 ug/m3, as cited by the European Environment Agency (EEA, 2024). The EU annual limit value for NO2 is 40 ug/m3 under Directive 2008/50/EC, which also sets sampling inlets between 1.5 m and 4 m above ground (EUR-Lex); the recast Directive (EU) 2024/2881 lowers it to 20 ug/m3 from 2030 (REHVA summary, checked 2026-10-24).
- FieldNode allows 115 mW average for sensors at its 5-day autonomy ceiling, with 100 mW proposed as its published design value (FND-CAL-001); AirStreet fits either.
- `budget_usd` in `project.yaml` is $285 for the sensor head (AST-DDR-002).
- PR-001 and PR-002 targets are proposals for review, set with reference to published field results; they are not taken from a standard.
- PR-017 and PR-018 were added on 2026-10-24 so that UR-014 and UR-015 each have a product requirement; their text is proposed, awaiting Amish.